A Mizar article in the style of Formalized Mathematics

On the Additive Chromatic Number of Several Families of Graphs

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Summary. Given a graph G, a labeling f of its vertices with natural numbers is an additive coloring of G if the sums $f(N(u))$ and $f(N(v))$ of the labels in the open neighborhoods of u and v differ for every edge uv of G. The additive chromatic number $\eta(G)$ is the least k such that G has an additive coloring with labels in $\{1, \ldots, k\}$ [3]. We formalize Sections 1–5 of [11], including the proofs of its propositions that are given in the appendix of its arXiv version: $\eta(G)$ of regular bipartite, complete multipartite and complete split graphs, of joins with complete graphs, fans, windmills, wheels, odd cycles, headless spiders and three families of suns, and the Additive Coloring Conjecture $\eta(G) \leq \chi(G)$ for complete multipartite graphs, for joins with complete graphs and for split graphs. The two results that the paper cites from [5] (the value of the topological additive number of complete multipartite digraphs and a lower bound from cliques), and the equality $\eta(G) = \min \eta_t(D)$ over the acyclic orientations D of G, are proved as well.

MSC: 05C15 05C78 68V20

Keywords: additive coloring; lucky labeling; additive chromatic number; split graph; complete multipartite graph

MML identifier: `ACN`, version: `8.1.15 5.94.1493`

INTRODUCTION

The Additive Coloring Problem asks for a labeling $f : V(G) \to \{1, \dots, k\}$ such that $f(N(u)) \neq f(N(v))$ for every edge uv, with k as small as possible; the least such k is the additive chromatic number (or lucky number) $\eta(G)$ [3]. The Additive Coloring Conjecture states that $\eta(G) \leq \chi(G)$ for every graph G. This article formalizes the paper [11] in the Mizar system [2, 4]. The proofs follow an earlier formalization of the same paper in Isabelle/HOL.

Graphs are finite simple graphs given as non empty, finite, irreflexive and symmetric relational structures [6]; Ω_G is the set of vertices of G and E_G its (symmetric) adjacency relation. The chromatic number $\chi(G)$ is the least number of classes of a partition of Ω_G into stable sets, as defined in [10, 9]. Sums of labels over finite sets are sums of bags [8, 7]. The additive chromatic number is introduced in Section 2, together with acyclic orientations (based on the reduction sequences of [1]), topological additive numberings, the topological additive number $\eta_t(D)$ of [5] (an extended real, $+\infty$ when there is no numbering) and the equality $\eta(G) = \min \eta_t(D)$ (32).

Each family of graphs of [11] is introduced as a predicate on graphs, with a theorem stating that such graphs exist. The results of the paper correspond to the following theorems: Observation 1 (40), Lemma 1 (41), Corollary 1 (42), Lemma 2 (66), (69), Theorem 1 (74), (76), Observation 2 (79), Theorem 2 (87), (89), fans (100), windmills (106), Proposition 1 (117), wheels (118), Theorem 3 (123), (124), (125), Proposition 2 (127), Lemma 3 (130), Corollary 2 (131), Proposition 3 (153), (156), Proposition 4 (198) and Proposition 5 (175), (181). The labeling of complete suns is the one of the appendix of [11], with one correction: in the case $m \equiv 2 \pmod 6$, $i = \lfloor m/2 \rfloor + 1$, the label of v_{i-1} is 2.

Throughout, $N(v)$ is the open neighborhood of v, $\deg v$ its degree, $f(N(v))$ the sum of the labels in $N(v)$, $\overline{\overline{X}}$ the cardinality of X, $\lceil a/b \rceil$ the ceiling of a/b, and $\mathrm{succ}_m(i)$, $\mathrm{pred}_m(i)$ the successor and the predecessor of i modulo m in $\{1, \dots, m\}$. In the families on two sides (spiders and suns), $u_i = \langle 0, i \rangle$ and $v_i = \langle 1, i \rangle$.

1. PRELIMINARIES ON SUMS OF LABELS

From now on I denotes a set.

From now on f, g denote functions from I into $\mathbb{N}$.

From now on J, K denote finite subsets of I.

From now on i, j, k, n, m denote natural numbers. Now we state the propositions:

(1) Let us consider a finite subset J of I. Suppose $J = \emptyset$. Then $\sum(f \restriction J) = 0$.

(2) Let us consider an object x. Suppose $J = \{x\}$. Then $\sum(f \restriction J) = f(x)$.

(3) If J misses K, then $\sum(f \restriction (J \cup K)) = \sum(f \restriction J) + \sum(f \restriction K)$.

(4) Let us consider an object x, and K. Suppose $x \in I$ and $x \notin J$ and $K = J \cup \{x\}$. Then $\sum(f \restriction K) = \sum(f \restriction J) + f(x)$.

(5) If (for every object x such that $x \in J$ holds $f(x) = g(x)$), then $\sum(f \restriction J) = \sum(g \restriction J)$.

The scheme *SumInd* deals with a set $\mathcal{I}$, a finite subset $\mathcal{J}$ of $\mathcal{I}$ and a unary predicate $\mathcal{P}$, and states that $\mathcal{P}[\mathcal{J}]$ provided $\mathcal{P}[\emptyset]$ and for every object x and every finite subset B of $\mathcal{I}$ such that $x \in \mathcal{J}$, $B \subseteq \mathcal{J}$, $x \notin B$ and $\mathcal{P}[B]$ holds $\mathcal{P}[B \cup \{x\}]$. Now we state the propositions:

(6) If (for every object x such that $x \in J$ holds $f(x) \leq g(x)$), then $\sum(f \restriction J) \leq \sum(g \restriction J)$.

(7) Let us consider natural numbers a, b. Suppose for every object x such that $x \in J$ holds $a \leq f(x)$ and $f(x) \leq b$. Then $a \cdot \overline{\overline{J}} \leq \sum(f \restriction J)$ and $\sum(f \restriction J) \leq b \cdot \overline{\overline{J}}$.

(8) Let us consider a natural number c. Suppose for every object x such that $x \in J$ holds $f(x) = c$. Then $\sum(f \restriction J) = c \cdot \overline{\overline{J}}$. The theorem is a consequence of (7).

(9) If $J \subseteq K$, then $\sum(f \restriction K) = \sum(f \restriction J) + \sum(f \restriction (K \setminus J))$. The theorem is a consequence of (3).

(10) If $J \subseteq K$, then $\sum(f \restriction J) \leq \sum(f \restriction K)$. The theorem is a consequence of (9).

(11) Let us consider an object x. Suppose $x \in J$. Then $f(x) \leq \sum(f \restriction J)$. The theorem is a consequence of (10) and (2).

(12) Let us consider objects x, y. Suppose $x \neq y$ and $J = \{x, y\}$. Then $\sum(f \restriction J) = f(x) + f(y)$. The theorem is a consequence of (4) and (2).

From now on h denotes a function from I into $\mathbb{N}$. Now we state the propositions:

(13) If (for every object x such that $x \in I$ holds $f(x) = 2^{h(x)}$) and h is one-to-one, then for every natural number m and every finite subset J of I such that (for every object x such that $x \in J$ holds $h(x) < m$) holds $\sum(f \restriction J) < 2^m$.

(14) If (for every object x such that $x \in I$ holds $f(x) = 2^{h(x)}$) and h is one-to-one, then for every J, K such that $\sum(f \restriction J) = \sum(f \restriction K)$ holds $J = K$.

(15) Let us consider a finite subset X of $\mathbb{N}$, and natural numbers a, b. Suppose $X \neq \emptyset$ and (for every natural number x such that $x \in X$ holds $a \leq x$ and $x \leq b$). Then $\overline{\overline{X}} + a \leq b + 1$.

(16) Let us consider natural numbers a, b. Suppose $J \neq \emptyset$ and (for every objects x, y such that $x \in J$ and $y \in J$ and $h(x) = h(y)$ holds $x = y$) and (for every object x such that $x \in J$ holds $a \leq h(x)$ and $h(x) \leq b$). Then $\overline{\overline{J}} + a \leq b + 1$.

2. GRAPHS, ADDITIVE COLORINGS AND THE ADDITIVE CHROMATIC NUMBER

A graph is a non empty finite irreflexive symmetric relational structure.

Let G be an irreflexive symmetric relational structure, and u, v be elements of G. We say that u and v are adjacent if and only if

(Def. 1) $\langle u, v \rangle \in E_G$.

From now on G denotes a graph.

From now on u, v, w denote elements of G.

Let G be a graph, and v be an element of G. The functor $N(v)$ yielding a subset of G is defined by the term

(Def. 2) $(E_G)^\circ\{v\}$.

Let G be a graph, and v be an element of G. One can check that $N(v)$ is finite.

Let G be a graph, and v be an element of G. The functor $\deg v$ yielding a natural number is defined by the term

(Def. 3) $\overline{\overline{N(v)}}$.

Now we state the propositions:

(17) $w \in N(v)$ if and only if v and w are adjacent.

(18) $v \notin N(v)$. The theorem is a consequence of (17).

(19) $w \in N(v)$ if and only if $v \in N(w)$. The theorem is a consequence of (17).

Let G be a graph, f be a function from Ω_G into $\mathbb{N}$, and v be an element of G. The functor $f(N(v))$ yielding an element of $\mathbb{N}$ is defined by the term

(Def. 4) $\sum(f{\restriction}N(v))$.

Let G be a relational structure, k be a natural number, and f be a function from Ω_G into $\mathbb{N}$. We say that f is k-labeling if and only if

(Def. 5) for every element v of G holds $1 \leq f(v)$ and $f(v) \leq k$.

Let G be a graph, and f be a function from Ω_G into $\mathbb{N}$. We say that f is additive if and only if

(Def. 6) for every u, v such that u and v are adjacent holds $f(N(u)) \neq f(N(v))$.

From now on f, g denote functions from Ω_G into $\mathbb{N}$. Now we state the propositions:

(20) If f is k-labeling and $k \leq n$, then f is n-labeling.

(21) There exist a natural number k and f such that f is k-labeling and additive.

Let G be a graph. The functor $\eta(G)$ yielding a natural number is defined by

(Def. 7) (there exists f such that f is it-labeling and additive) and for every natural number k and f such that f is k-labeling and additive holds $it \leq k$.

Now we state the propositions:

(22) If f is k-labeling and additive, then $\eta(G) \leq k$.

(23) There exists f such that f is $\eta(G)$-labeling and additive.

(24) $1 \leq \eta(G)$. The theorem is a consequence of (23).

(25) If (for every natural number k and f such that f is k-labeling and additive holds $n \leq k$), then $n \leq \eta(G)$. The theorem is a consequence of (23).

Let G be a graph, and A be a relation. We say that A is an orientation of G if and only if

(Def. 8) $A \subseteq E_G$ and (for every u, v such that u and v are adjacent holds $\langle u, v\rangle \in A$ or $\langle v, u\rangle \in A$) and for every objects x, y such that $\langle x, y\rangle \in A$ holds $\langle y, x\rangle \notin A$.

Let A be a relation. We say that A is acyclic if and only if

(Def. 9) for every reduction sequence p of A such that $p(1) = p(\operatorname{len} p)$ holds $\operatorname{len} p = 1$.

Now we state the proposition:

(26) Let us consider a relation A, and a natural-valued function s. Suppose for every objects x, y such that $\langle x, y\rangle \in A$ holds $s(x) < s(y)$. Then A is acyclic.

Let G be a graph, A be a relation, and f be a function from Ω_G into $\mathbb{N}$. We say that f is topological for A if and only if

(Def. 10) for every u, v such that $\langle u, v\rangle \in A$ holds $f(N(u)) < f(N(v))$.

Let G be a graph, and A be a relation. The functor $\mathrm{Num}(G, A)$ yielding a subset of $\mathbb{N}$ is defined by the term

(Def. 11) $\{k \in \mathbb{N}$: there exists f such that f is k-labeling and f is topological for $A\}$.

Let G be a graph, and A be a relation. The functor $\eta_t(G, A)$ yielding an extended real is defined by the term

(Def. 12) $\inf \mathrm{Num}(G, A)$.

From now on A denotes a relation. Now we state the propositions:

(27) If f is k-labeling and f is topological for A, then $\eta_t(G, A) \leq k$.

(28) If $\eta_t(G, A) \neq +\infty$, then there exist a natural number k and f such that $\eta_t(G, A) = k$ and f is k-labeling and f is topological for A.

(29) If A is an orientation of G and f is topological for A, then f is additive.

(30) If A is an orientation of G, then $\eta(G) \leq \eta_t(G, A)$.

(31) If f is additive, then there exists a relation A on Ω_G such that A is an orientation of G and A is acyclic and f is topological for A.

(32) (i) for every A such that A is an orientation of G and A is acyclic holds $\eta(G) \leq \eta_t(G, A)$, and
(ii) there exists A such that A is an orientation of G and A is acyclic and $\eta_t(G, A) = \eta(G)$.

The theorem is a consequence of (30), (23), (31), and (27).

(33) Let us consider a clique Q of G. Suppose $u \in Q$ and $v \in Q$ and $u \neq v$. Then u and v are adjacent.

(34) Let us consider a subset Q of G. Suppose for every u, v such that $u \in Q$ and $v \in Q$ and $u \neq v$ holds u and v are adjacent. Then Q is a clique of G.

(35) Let us consider a finite clique Q of G. Then $\overline{\overline{Q}} \leq \chi(G)$.

(36) Let us consider a natural number c. Suppose for every w such that $w \in N(v)$ holds $f(w) = c$. Then $f(N(v)) = c \cdot \deg v$. The theorem is a consequence of (8).

(37) If f is k-labeling, then $\deg v \leq f(N(v))$ and $f(N(v)) \leq k \cdot \deg v$. The theorem is a consequence of (7).

Let a, b be natural numbers. The functor $\lceil \frac{a}{b} \rceil$ yielding a natural number is defined by the term

(Def. 13) $(a + b \dot{-} 1) \operatorname{div} b$.

From now on a, b, c denote natural numbers. Now we state the propositions:

(38) If $0 < b$, then ($\lceil \frac{a}{b} \rceil \leq k$ if and only if $a \leq k \cdot b$).

(39) If $0 < b$, then $a \leq \lceil \frac{a}{b} \rceil \cdot b$. The theorem is a consequence of (38).

The scheme *LabEx* deals with a graph $\mathcal{G}$ and a unary functor $\mathcal{F}$ from the elements of $\mathcal{G}$ into the natural numbers, and states that there exists a function f from $\Omega_{\mathcal{G}}$ into $\mathbb{N}$ such that for every element v of $\mathcal{G}$, $f(v) = \mathcal{F}(v)$.

3. REGULAR BIPARTITE AND COMPLETE MULTIPARTITE GRAPHS

Now we state the propositions:

(40) $\eta(G) = 1$ if and only if for every u, v such that u and v are adjacent holds $\deg u \neq \deg v$.

(41) Let us consider subsets U, W of G. Suppose U misses W and $U \cup W = \Omega_G$ and (for every u, v such that u and v are adjacent holds ($u \in U$ if and only if $v \in W$)) and (for every u, v such that $v \in W$ and $u \in N(v)$ holds $\deg u < 2 \cdot \deg v$). Then

  (i) if (for every u, v such that u and v are adjacent holds $\deg u \neq \deg v$), then $\eta(G) = 1$, and

  (ii) if (there exist u, v such that u and v are adjacent and $\deg u = \deg v$), then $\eta(G) = 2$.

(42) Let us consider subsets U, W of G. Suppose U misses W and $U \cup W = \Omega_G$ and (for every u, v such that u and v are adjacent holds ($u \in U$ if and only if $v \in W$)) and (for every u, v holds $\deg u = \deg v$) and (there exist u, v such that u and v are adjacent). Then $\eta(G) = 2$.

(43) Let us consider a finite subset X of $\mathbb{N}$. Then there exists a natural number m such that for every natural number k such that $k \in X$ holds $k \leq m$. The theorem is a consequence of (11).

From now on n denotes a finite sequence of $\mathbb{N}$.

Let n be a finite sequence of $\mathbb{N}$, and i be a natural number. Assume that $1 \leq i$ and $i \leq \operatorname{len} n$. The functor $\mathrm{s}(n, i)$ yielding a natural number is defined by

(Def. 14) (there exists a natural number t such that $i \leq t$ and $t \leq \operatorname{len} n$ and $it + i = n(t) + t$) and for every natural number t such that $i \leq t$ and $t \leq \operatorname{len} n$ holds $n(t) + t \leq it + i$.

Now we state the propositions:

(44) Let us consider natural numbers i, x. Suppose $1 \leq i$ and $i \leq \operatorname{len} n$. Then $\mathrm{s}(n, i) \leq x$ if and only if for every natural number t such that $i \leq t$ and $t \leq \operatorname{len} n$ holds $n(t) + t \leq x + i$.

(45) Let us consider a natural number i. Suppose $1 \leq i$ and $i \leq \operatorname{len} n$. Then $n(i) \leq \mathrm{s}(n, i)$.

(46) If $1 \leq \operatorname{len} n$, then $\mathrm{s}(n, \operatorname{len} n) = n(\operatorname{len} n)$.

(47) Let us consider a natural number i. Suppose $1 \leq i$ and $i < \operatorname{len} n$. Then $\mathrm{s}(n, i) = \max(\mathrm{s}(n, i + 1) + 1, n(i))$.

(48) Let us consider natural numbers i, j. Suppose $1 \leq i$ and $i < j$ and $j \leq \operatorname{len} n$. Then $\mathrm{s}(n, j) + (j - i) \leq \mathrm{s}(n, i)$.

Let n be a finite sequence of $\mathbb{N}$. The functor $\operatorname{bound} n$ yielding a natural number is defined by

(Def. 15) (for every natural number i such that $1 \leq i$ and $i \leq \operatorname{len} n$ holds $\left\lceil \frac{\mathrm{s}(n,i)}{n(i)} \right\rceil \leq it$) and (if $1 \leq \operatorname{len} n$, then there exists a natural number i such that $1 \leq i$ and $i \leq \operatorname{len} n$ and $it = \left\lceil \frac{\mathrm{s}(n,i)}{n(i)} \right\rceil$) and (if $\operatorname{len} n = 0$, then $it = 0$).

Now we state the propositions:

(49) If (for every natural number i such that $1 \leq i$ and $i \leq \operatorname{len} n$ holds $1 \leq n(i)$), then for every natural number K holds $\operatorname{bound} n \leq K$ if and only if for every natural numbers i, t such that $1 \leq i$ and $i \leq t$ and $t \leq \operatorname{len} n$ holds $n(t) + t \leq K \cdot n(i) + i$.

(50) If (for every natural number i such that $1 \leq i$ and $i \leq \operatorname{len} n$ holds $1 \leq n(i)$), then for every natural number i such that $1 \leq i$ and $i \leq \operatorname{len} n$ holds $\mathrm{s}(n, i) \leq \operatorname{bound} n \cdot n(i)$. The theorem is a consequence of (38).

From now on P, Q denote finite sequences of 2^{Ω_G}.

Let G be a graph, and P be a finite sequence of 2^{Ω_G}. We say that P is a complete multipartition of G if and only if

(Def. 16) (for every i such that $i \in \operatorname{dom} P$ holds $P(i) \neq \emptyset$) and (for every i, j such that $i \in \operatorname{dom} P$ and $j \in \operatorname{dom} P$ and $i \neq j$ holds $P(i)$ misses $P(j)$) and (for every v holds there exists i such that $i \in \operatorname{dom} P$ and $v \in P(i)$) and for every u, v holds u and v are adjacent iff there exist i, j such that $i \in \operatorname{dom} P$ and $j \in \operatorname{dom} P$ and $i \neq j$ and $u \in P(i)$ and $v \in P(j)$.

Now we state the propositions:

(51) If $i \in \operatorname{dom} P$, then $P(i) \subseteq \Omega_G$.

(52) If P is a complete multipartition of G and $i \in \operatorname{dom} P$ and $j \in \operatorname{dom} P$ and $v \in P(i)$ and $v \in P(j)$, then $i = j$.

Let G be a graph, and P be a finite sequence of 2^{Ω_G}. The functor $\operatorname{sizes} P$ yielding a finite sequence of $\mathbb{N}$ is defined by

(Def. 17) $\operatorname{len} it = \operatorname{len} P$ and for every i such that $i \in \operatorname{dom} P$ holds $it(i) = \overline{\overline{P(i)}}$.

Now we state the propositions:

(53) If P is a complete multipartition of G, then for every i such that $1 \leq i$ and $i \leq \operatorname{len} (\operatorname{sizes} P)$ holds $1 \leq (\operatorname{sizes} P)(i)$.

(54) If P is a complete multipartition of G and $j \in \operatorname{dom} P$ and $w \in P(j)$, then $N(w) = (\Omega_G) \setminus P(j)$.

(55) If P is a complete multipartition of G and $j \in \operatorname{dom} P$ and $w \in P(j)$, then for every finite subset J of Ω_G such that $J = P(j)$ holds $f(N(w)) + \sum(f \restriction J) = \sum(f \restriction \Omega_G)$. The theorem is a consequence of (54) and (9).

(56) $P(i) \subseteq \Omega_G$. The theorem is a consequence of (51).

Let G be a graph, f be a function from Ω_G into $\mathbb{N}$, P be a finite sequence of 2^{Ω_G}, and i be a natural number. The functor $f(P(i))$ yielding an element of $\mathbb{N}$ is defined by

(Def. 18) for every finite subset J of Ω_G such that $J = P(i)$ holds $it = \sum(f \restriction J)$.

Now we state the propositions:

(57) If f is k-labeling and $i \in \operatorname{dom} P$, then $(\operatorname{sizes} P)(i) \leq f(P(i))$ and $f(P(i)) \leq k \cdot (\operatorname{sizes} P)(i)$. The theorem is a consequence of (56) and (7).

(58) If P is a complete multipartition of G and $j \in \operatorname{dom} P$ and $w \in P(j)$, then $f(N(w)) + f(P(j)) = \sum(f \restriction \Omega_G)$. The theorem is a consequence of (56) and (55).

Let G be a graph, P be a finite sequence of 2^{Ω_G}, and A be a relation. We say that A is monotone for P if and only if

(Def. 19) for every objects x, y such that $\langle x, y\rangle \in A$ holds there exist i, j such that $i \in \operatorname{dom} P$ and $j \in \operatorname{dom} P$ and $i < j$ and $x \in P(i)$ and $y \in P(j)$.

Now we state the propositions:

(59) If P is a complete multipartition of G and A is an orientation of G and A is monotone for P and $i \in \operatorname{dom} P$ and $j \in \operatorname{dom} P$ and $i < j$ and $u \in P(i)$ and $w \in P(j)$, then $\langle u, w\rangle \in A$. The theorem is a consequence of (52).

(60) Let us consider a natural-valued function *key*, a natural number r, and natural numbers i, t. Suppose for every natural numbers i, j such that $1 \leq i$ and $i < j$ and $j \leq r$ holds $\mathit{key}(j) < \mathit{key}(i)$ and $1 \leq i$ and $i \leq t$ and $t \leq r$. Then $\mathit{key}(t) + t \leq \mathit{key}(i) + i$.

(61) If P is a complete multipartition of G and $i \in \operatorname{dom} P$, then there exists u such that $u \in P(i)$. The theorem is a consequence of (56).

(62) If P is a complete multipartition of G and A is an orientation of G and A is monotone for P and f is topological for A and $i \in \operatorname{dom} P$ and $j \in \operatorname{dom} P$ and $i < j$, then $f(P(j)) < f(P(i))$.

(63) If P is a complete multipartition of G and A is an orientation of G and A is monotone for P and f is k-labeling and f is topological for A, then $\operatorname{bound}(\operatorname{sizes} P) \leq k$.

(64) Let us consider a set I, a finite subset S of I, natural numbers K, T, and a function h from I into $\mathbb{N}$. Suppose $\overline{\overline{S}} \leq T$ and $T \leq K \cdot \overline{\overline{S}}$. Then there exists a function g from I into $\mathbb{N}$ such that (for every object x such that $x \in S$ holds $1 \leq g(x)$ and $g(x) \leq K$) and (for every object x such that $x \in I$ and $x \notin S$ holds $g(x) = h(x)$) and $\sum(g \restriction S) = T$.

(65) If P is a complete multipartition of G and A is monotone for P, then there exists f such that f is $\operatorname{bound}(\operatorname{sizes} P)$-labeling and f is topological for A.

(66) If P is a complete multipartition of G and A is an orientation of G and A is monotone for P, then $\eta_t(G, A) = \operatorname{bound}(\operatorname{sizes} P)$.

(67) Let us consider finite sets X, Y. Suppose $X \subseteq Y$ and $\overline{\overline{X}} = \overline{\overline{Y}}$. Then $X = Y$.

(68) Let us consider finite sets X, Y, and an object y. Suppose $X \subseteq Y$ and $y \in Y$ and $y \notin X$. Then $\overline{\overline{X}} < \overline{\overline{Y}}$.

(69) If P is a complete multipartition of G and A is an orientation of G and A is acyclic, then ($\eta_t(G, A) \neq +\infty$ if and only if there exists Q such that Q is a complete multipartition of G and A is monotone for Q).

(70) Let us consider a finite subset X of $\mathbb{N}$. Suppose $X \neq \emptyset$. Then there exists a natural number M such that $M \in X$ and for every natural number k such that $k \in X$ holds $k \leq M$.

(71) Let us consider a subset X of $\mathbb{N}$. Suppose $X \neq \emptyset$. Then there exists a natural number m such that $m \in X$ and for every natural number k such that $k \in X$ holds $m \leq k$.

(72) Let us consider natural numbers i, t. Suppose $1 \leq i$ and $i \leq t$. Then

(i) $\overline{\overline{\mathrm{Seg}\, t \setminus \mathrm{Seg}\, (i \dot{-} 1)}} = t + 1 - i$, and

(ii) for every natural number l holds $l \in \mathrm{Seg}\, t \setminus \mathrm{Seg}\, (i \dot{-} 1)$ if and only if $i \leq l$ and $l \leq t$.

(73) If P is a complete multipartition of G, then there exists a relation A on Ω_G such that A is an orientation of G and A is acyclic and A is monotone for P.

(74) If P is a complete multipartition of G and (for every i, j such that $1 \leq i$ and $i \leq j$ and $j \leq \mathrm{len}\, P$ holds $(\mathrm{sizes}\, P)(j) \leq (\mathrm{sizes}\, P)(i)$), then $\eta(G) = \mathrm{bound}\, (\mathrm{sizes}\, P)$ and $\eta(G) \leq \mathrm{len}\, P$.

(75) If P is a complete multipartition of G, then $\mathrm{len}\, P \leq \chi(G)$.

(76) If P is a complete multipartition of G and (for every i, j such that $1 \leq i$ and $i \leq j$ and $j \leq \mathrm{len}\, P$ holds $(\mathrm{sizes}\, P)(j) \leq (\mathrm{sizes}\, P)(i)$), then $\eta(G) \leq \chi(G)$. The theorem is a consequence of (74) and (75).

4. JOINS WITH COMPLETE GRAPHS

Let G be a graph, and u, v be elements of G. We say that u and v are true twins if and only if

(Def. 20) $N(u) \cup \{u\} = N(v) \cup \{v\}$.

Now we state the propositions:

(77) Let us consider a finite subset J of Ω_G. Suppose $J = N(v) \cup \{v\}$. Then $\sum(f \restriction J) = f(N(v)) + f(v)$. The theorem is a consequence of (18) and (4).

(78) If f is additive and u and v are true twins and $u \neq v$, then $f(u) \neq f(v)$.

(79) Let us consider a subset T of G. Suppose for every u, v such that $u \in T$ and $v \in T$ holds u and v are true twins. Then $\overline{\overline{T}} \leq \eta(G)$.

Let G be a graph. The functor $\Delta(G)$ yielding a natural number is defined by

(Def. 21) (there exists v such that $\mathit{it} = \deg v$) and for every v holds $\deg v \leq \mathit{it}$.

From now on H denotes a graph.

Let G, H be graphs, and U be a set. We say that H is a join of G and U if and only if

(Def. 22) $\Omega_H = (\Omega_G) \cup U$ and U misses Ω_G and for every objects x, y holds $\langle x, y \rangle \in E_H$ iff $\langle x, y \rangle \in E_G$ or ($x \in U$ and $y \in U$ and $x \neq y$) or ($x \in \Omega_G$ and $y \in U$) or ($x \in U$ and $y \in \Omega_G$).

Now we state the propositions:

(80) Let us consider a finite set U. Suppose U misses Ω_G. Then there exists a strict graph H such that H is a join of G and U.

(81) Let us consider sets I_1, I_2, a function f_1 from I_1 into $\mathbb{N}$, a function f_2 from I_2 into $\mathbb{N}$, a finite subset J_1 of I_1, and a finite subset J_2 of I_2. Suppose $J_1 = J_2$ and for every object x such that $x \in J_1$ holds $f_1(x) = f_2(x)$. Then $\sum(f_1 \restriction J_1) = \sum(f_2 \restriction J_2)$.

From now on U denotes a finite set. Now we state the propositions:

(82) If H is a join of G and U, then for every elements x, y of H such that $x = v$ and $y = w$ holds x and y are adjacent if and only if v and w are adjacent.

(83) If H is a join of G and U, then for every element x of H such that $x = v$ holds $N(x) = N(v) \cup U$.

(84) If H is a join of G and U, then for every element x of H such that $x \in U$ holds $N(x) = (U \setminus \{x\}) \cup \Omega_G$.

From now on F denotes a function from Ω_H into $\mathbb{N}$. Now we state the propositions:

(85) If H is a join of G and U, then for every element x of H and every finite subsets NG, UU of Ω_H such that $x = v$ and $NG = N(v)$ and $UU = U$ holds $F(N(x)) = \sum(F \upharpoonright NG) + \sum(F \upharpoonright UU)$. The theorem is a consequence of (83) and (3).

(86) If H is a join of G and U, then for every element x of H and every finite subsets UU, VV of Ω_H such that $x \in U$ and $UU = U$ and $VV = \Omega_G$ holds $F(N(x)) + F(x) = \sum(F \upharpoonright UU) + \sum(F \upharpoonright VV)$.

(87) If H is a join of G and U and $\overline{\overline{U}} + \Delta(G) + 1 \leq \overline{\overline{\Omega_G}}$, then $\eta(H) = \max(\eta(G), \overline{\overline{U}})$.

(88) If H is a join of G and U, then $\chi(G) + \overline{\overline{U}} \leq \chi(H)$.

(89) If H is a join of G and U and $\overline{\overline{U}} + \Delta(G) + 1 \leq \overline{\overline{\Omega_G}}$ and $\eta(G) \leq \chi(G)$, then $\eta(H) \leq \chi(H)$. The theorem is a consequence of (87) and (88).

The scheme *GraphEx* deals with a non empty finite set $\mathcal{V}$ and a binary predicate $\mathcal{P}$, and states that there exists a strict graph G such that $\Omega_G = \mathcal{V}$ and for all objects x, y, $\langle x, y \rangle \in E_G$ iff $x \in \mathcal{V}$, $y \in \mathcal{V}$ and $\mathcal{P}[x, y]$, provided for all objects x, y such that $\mathcal{P}[x, y]$ holds $\mathcal{P}[y, x]$, and for every object x not $\mathcal{P}[x, x]$. Now we state the propositions:

(90) Let us consider natural numbers i, j. Suppose $u = i$ and $v = j$. Then u and v are adjacent if and only if $\langle i, j \rangle \in E_G$.

(91) If u and v are adjacent and $\deg u = \deg v$, then $2 \leq \eta(G)$. The theorem is a consequence of (40) and (24).

(92) If u and v are adjacent and $\deg u = \deg v$ and f is 2-labeling and additive, then $\eta(G) = 2$. The theorem is a consequence of (91) and (22).

5. Paths and Fans

Let G be a graph, and N be a natural number. We say that G is a path of order N if and only if

(Def. 23) $\Omega_G = \operatorname{Seg} N$ and for every natural numbers i, j holds $\langle i, j \rangle \in E_G$ iff $i \in \operatorname{Seg} N$ and $j \in \operatorname{Seg} N$ and ($j = i + 1$ or $i = j + 1$).

Now we state the propositions:

(93) Let us consider a natural number N. Suppose $1 \leq N$. Then there exists a strict graph G such that G is a path of order N.

(94) Let us consider a natural number N, and a natural number i. Suppose G is a path of order N and $2 \leq N$ and $v = i$. Then

(i) if $i = 1$, then $N(v) = \{2\}$, and

(ii) if $1 < i$ and $i < N$, then $N(v) = \{i \dot{-} 1, i + 1\}$, and
(iii) if $i = N$, then $N(v) = \{N \dot{-} 1\}$.

(95) Let us consider a natural number N, and a natural number i. Suppose G is a path of order N and $2 \leq N$ and $v = i$. Then
(i) if $i = 1$, then $f(N(v)) = f(2)$, and
(ii) if $1 < i$ and $i < N$, then $f(N(v)) = f(i \dot{-} 1) + f(i + 1)$, and
(iii) if $i = N$, then $f(N(v)) = f(N \dot{-} 1)$.

(96) Let us consider a natural number N, and a natural number i. Suppose G is a path of order N and $2 \leq N$ and $v = i$ and $1 < i$ and $i < N$. Then $\deg v = 2$. The theorem is a consequence of (94).

(97) Let us consider a natural number N. Suppose G is a path of order N and $4 \leq N$. Then $\eta(G) = 2$.

(98) Let us consider a natural number N. Suppose G is a path of order N and $2 \leq N$. Then $\Delta(G) \leq 2$.

(99) Let us consider an object w. Suppose H is a join of G and $\{w\}$ and $\Delta(G)+2 \leq \overline{\overline{\Omega_G}}$. Then $\eta(H) = \eta(G)$. The theorem is a consequence of (87) and (24).

(100) Let us consider a natural number N, and an object w. Suppose $4 \leq N$ and G is a path of order N and H is a join of G and $\{w\}$. Then $\eta(H) = 2$. The theorem is a consequence of (98), (99), and (97).

6. WINDMILLS

Let G be a graph, and m, p be natural numbers. We say that G is m copies of K_p if and only if

(Def. 24) $\Omega_G = \operatorname{Seg} m \times \operatorname{Seg} p$ and for every natural numbers i, j, i_1, j_1 holds $\langle\langle i, j\rangle, \langle i_1, j_1\rangle\rangle \in E_G$ iff $i \in \operatorname{Seg} m$ and $j \in \operatorname{Seg} p$ and $j_1 \in \operatorname{Seg} p$ and $i_1 = i$ and $j_1 \neq j$.

Now we state the propositions:

(101) Let us consider natural numbers m, p. Suppose $1 \leq m$ and $1 \leq p$. Then there exists a strict graph G such that G is m copies of K_p.

(102) Let us consider natural numbers m, p. Suppose G is m copies of K_p. Then there exist natural numbers i, j such that $v = \langle i, j\rangle$ and $i \in \operatorname{Seg} m$ and $j \in \operatorname{Seg} p$.

(103) Let us consider natural numbers m, p, and natural numbers i, j. Suppose G is m copies of K_p and $v = \langle i, j\rangle$. Then $N(v) \cup \{v\} = \{i\} \times \operatorname{Seg} p$.

(104) Let us consider natural numbers m, p. Suppose G is m copies of K_p. Then $\deg v + 1 = p$. The theorem is a consequence of (102), (103), and (18).

(105) Let us consider natural numbers m, p. Suppose G is m copies of K_p. Then $\eta(G) = p$.

(106) Let us consider natural numbers m, p, and an object w. Suppose $2 \leq m$ and $2 \leq p$ and G is m copies of K_p and H is a join of G and $\{w\}$. Then $\eta(H) = p$.

7. CYCLES AND WHEELS

Let N, i be natural numbers. The functor $\mathrm{succ}_N(i)$ yielding a natural number is defined by the term

(Def. 25) $\begin{cases} i+1, & \text{if } i < N, \\ 1, & \text{otherwise.} \end{cases}$

Let N, i be natural numbers. The functor $\mathrm{pred}_N(i)$ yielding a natural number is defined by the term

(Def. 26) $\begin{cases} i \dot{-} 1, & \text{if } 1 < i, \\ N, & \text{otherwise.} \end{cases}$

Now we state the propositions:

(107) Let us consider natural numbers N, i. Suppose $i \in \mathrm{Seg}\, N$. Then $\mathrm{succ}_N(i) \in \mathrm{Seg}\, N$.

(108) Let us consider natural numbers N, i. Suppose $i \in \mathrm{Seg}\, N$. Then $\mathrm{pred}_N(i) \in \mathrm{Seg}\, N$.

(109) Let us consider natural numbers N, i. Suppose $i \in \mathrm{Seg}\, N$. Then $\mathrm{pred}_N(\mathrm{succ}_N(i)) = i$ and $\mathrm{succ}_N(\mathrm{pred}_N(i)) = i$.

Let G be a graph, and N be a natural number. We say that G is a cycle of order N if and only if

(Def. 27) $\Omega_G = \mathrm{Seg}\, N$ and for every natural numbers i, j holds $\langle i, j \rangle \in E_G$ iff $i \in \mathrm{Seg}\, N$ and $j \in \mathrm{Seg}\, N$ and ($j = \mathrm{succ}_N(i)$ or $i = \mathrm{succ}_N(j)$).

Now we state the propositions:

(110) Let us consider a natural number N. Suppose $3 \leq N$. Then there exists a strict graph G such that G is a cycle of order N.

(111) Let us consider a natural number N, and a natural number i. Suppose G is a cycle of order N and $v = i$. Then $N(v) = \{\mathrm{succ}_N(i), \mathrm{pred}_N(i)\}$.

(112) Let us consider natural numbers N, i. Suppose $3 \leq N$ and $i \in \mathrm{Seg}\, N$. Then $\mathrm{succ}_N(i) \neq \mathrm{pred}_N(i)$.

(113) Let us consider a natural number N, and a natural number i. Suppose $3 \leq N$ and G is a cycle of order N and $v = i$. Then $f(N(v)) = f(\mathrm{succ}_N(i)) + f(\mathrm{pred}_N(i))$ and $\deg v = 2$. The theorem is a consequence of (111), (112), and (12).

(114) Let us consider natural numbers N, i. Suppose N is even and $i \in \mathrm{Seg}\, N$. Then ($\mathrm{succ}_N(i)$ is odd if and only if i is even).

(115) Let us consider a natural number N. Suppose $3 \leq N$ and N is even and G is a cycle of order N. Then $\eta(G) = 2$.

(116) Let us consider a natural number N. Suppose $3 \leq N$ and N is odd and G is a cycle of order N. Then $3 \leq \eta(G)$.

(117) Let us consider a natural number N. Suppose $5 \leq N$ and N is odd and G is a cycle of order N. Then $\eta(G) = 3$.

(118) Let us consider a natural number N, and an object w. Suppose $4 \leq N$ and G is a cycle of order N and H is a join of G and $\{w\}$. Then
    (i) if N is even, then $\eta(H) = 2$, and
    (ii) if N is odd, then $\eta(H) = 3$.

8. Split Graphs

From now on Q_1, S_1, T denote subsets of G.

Let G be a graph, and Q_1, S_1 be subsets of G. We say that G is split by Q_1 and S_1 if and only if

(Def. 28) $Q_1 \cup S_1 = \Omega_G$ and Q_1 misses S_1 and (for every u, v such that $u \in Q_1$ and $v \in Q_1$ and $u \neq v$ holds u and v are adjacent) and (for every u, v such that $u \in S_1$ and $v \in S_1$ holds u and v are not adjacent) and for every v such that $v \in S_1$ holds there exists u such that $u \in Q_1$ and u and v are not adjacent.

Now we state the propositions:

(119) If G is split by Q_1 and S_1 and $v \in S_1$, then $N(v) \subseteq Q_1$. The theorem is a consequence of (17).

(120) If G is split by Q_1 and S_1 and $u \in Q_1$, then $N(u) = (Q_1 \setminus \{u\}) \cup (N(u) \cap S_1)$ and $(Q_1 \setminus \{u\})$ misses $(N(u) \cap S_1)$.

(121) If G is split by Q_1 and S_1 and $u \in Q_1$, then $\deg u + 1 = \overline{\overline{Q_1}} + \overline{\overline{N(u) \cap S_1}}$. The theorem is a consequence of (120).

(122) If G is split by Q_1 and S_1 and $u \in Q_1$, then for every natural number c such that for every w such that $w \in S_1$ holds $f(w) = c$ holds $f(N(u)) + f(u) = \sum (f \restriction Q_1) + c \cdot \overline{\overline{N(u) \cap S_1}}$.

(123) If G is split by Q_1 and S_1 and $T \subseteq Q_1$ and $T \neq \emptyset$ and (for every u, w such that $u \in T$ and $w \in T$ and $\deg u = \deg w$ holds $u = w$), then $\eta(G) \leq \overline{\overline{Q_1}} - \overline{\overline{T}} + 1$.

(124) If G is split by Q_1 and S_1, then $\eta(G) \leq \overline{\overline{Q_1}}$ and $\eta(G) \leq \chi(G)$.

(125) If $Q_1 \cup S_1 = \Omega_G$ and Q_1 misses S_1 and (for every u, v such that $u \in Q_1$ and $v \in Q_1$ and $u \neq v$ holds u and v are adjacent) and (for every u, v such that $u \in S_1$ and $v \in S_1$ holds u and v are not adjacent), then $\eta(G) \leq \chi(G)$.

(126) If (for every u such that $u \neq w$ holds w and u are adjacent), then $N(w) \cup \{w\} = \Omega_G$.

(127) If $Q_1 \cup S_1 = \Omega_G$ and Q_1 misses S_1 and (for every u, v such that $u \in Q_1$ and $v \in Q_1$ and $u \neq v$ holds u and v are adjacent) and (for every u, v such that $u \in S_1$ and $v \in S_1$ holds u and v are not adjacent) and (for every u, v such that $u \in Q_1$ and $v \in S_1$ holds u and v are adjacent) and $Q_1 \neq \emptyset$ and $2 \leq \overline{\overline{S_1}}$, then $\eta(G) = \overline{\overline{Q_1}}$.

(128) Let us consider a finite clique Q of G. Suppose $u \in Q$. Then $N(u) \cap Q = Q \setminus \{u\}$ and $\overline{\overline{Q}} \leq \deg u + 1$.

(129) Let us consider natural numbers a, a_1, b, b_1. Suppose $a \leq a_1$ and $0 < b_1$ and $b_1 \leq b$. Then $\lceil \frac{a}{b} \rceil \leq \left\lceil \frac{a_1}{b_1} \right\rceil$. The theorem is a consequence of (39) and (38).

(130) Let us consider a finite clique Q of G, and elements q_F, q_L of G. Suppose A is an orientation of G and $q_F \in Q$ and $q_L \in Q$ and (for every element q of G such that $q \in Q$ and $q \neq q_F$ holds $\langle q_F, q \rangle \in A$) and (for every element q of G such that $q \in Q$ and $q \neq q_L$ holds $\langle q, q_L \rangle \in A$). Then $\left\lceil \frac{\deg q_F + 1}{\deg q_L + 2 \dot{-} \overline{\overline{Q}}} \right\rceil \leq \eta_t(G, A)$.

(131) Let us consider a finite clique Q of G, and natural numbers d_1, d_2. Suppose $Q \neq \emptyset$ and for every element q of G such that $q \in Q$ holds $d_1 \leq \deg q$ and $\deg q \leq d_2$. Then $\left\lceil \frac{d_1+1}{d_2+2 \mathbin{\dot{-}} \overline{\overline{Q}}} \right\rceil \leq \eta(G)$.

9. HEADLESS SPIDERS

Now we state the propositions:

(132) Let us consider a set C, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose $C \subseteq \{0,1\} \times \mathbb{N}$. Then there exists a function f from C into $\mathbb{N}$ such that for every natural number i holds (if $\langle 0, i \rangle \in C$, then $f(\langle 0, i \rangle) = f_u(i)$) and (if $\langle 1, i \rangle \in C$, then $f(\langle 1, i \rangle) = f_v(i)$).

(133) Let us consider a set I, a function f from I into $\mathbb{N}$, an object a, a finite subset X of $\mathbb{N}$, a function g from $\mathbb{N}$ into $\mathbb{N}$, and a finite subset J of I. Suppose $\{a\} \times X \subseteq I$ and (for every natural number j such that $j \in X$ holds $f(\langle a, j \rangle) = g(j)$) and $J = \{a\} \times X$. Then $\sum (f {\restriction} J) = \sum (g {\restriction} X)$.

(134) Let us consider a set I, a function f from I into $\mathbb{N}$, finite subsets A, B of $\mathbb{N}$, functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$, and a finite subset J of I. Suppose $\{0\} \times A \subseteq I$ and $\{1\} \times B \subseteq I$ and (for every natural number j such that $j \in A$ holds $f(\langle 0, j \rangle) = f_u(j)$) and (for every natural number j such that $j \in B$ holds $f(\langle 1, j \rangle) = f_v(j)$) and $J = \{0\} \times A \cup \{1\} \times B$. Then $\sum (f {\restriction} J) = \sum (f_u {\restriction} A) + \sum (f_v {\restriction} B)$.

(135) Let us consider an object z. Then $z \in N(v)$ if and only if $\langle v, z \rangle \in E_G$.

(136) Let us consider a natural number q. Suppose $\Omega_G = \{0,1\} \times \operatorname{Seg} q$. Then there exist functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$ such that for every natural number j such that $j \in \operatorname{Seg} q$ holds $f_u(j) = f(\langle 0, j \rangle)$ and $f_v(j) = f(\langle 1, j \rangle)$.

Let G be a graph, and q be a natural number. We say that G is a thin spider of order q if and only if

(Def. 29) $\Omega_G = \{0,1\} \times \operatorname{Seg} q$ and for every objects x, y holds $\langle x, y \rangle \in E_G$ iff there exist natural numbers i, j such that $i \in \operatorname{Seg} q$ and $j \in \operatorname{Seg} q$ and (($x = \langle 0, i \rangle$ and $y = \langle 0, j \rangle$ and $i \neq j$) or ($x = \langle 0, i \rangle$ and $y = \langle 1, j \rangle$ and $i = j$) or ($x = \langle 1, i \rangle$ and $y = \langle 0, j \rangle$ and $i = j$)).

Let G be a graph, and q be a natural number. We say that G is a thick spider of order q if and only if

(Def. 30) $\Omega_G = \{0,1\} \times \operatorname{Seg} q$ and for every objects x, y holds $\langle x, y \rangle \in E_G$ iff there exist natural numbers i, j such that $i \in \operatorname{Seg} q$ and $j \in \operatorname{Seg} q$ and (($x = \langle 0, i \rangle$ and $y = \langle 0, j \rangle$ and $i \neq j$) or ($x = \langle 0, i \rangle$ and $y = \langle 1, j \rangle$ and $i \neq j$) or ($x = \langle 1, i \rangle$ and $y = \langle 0, j \rangle$ and $i \neq j$)).

Now we state the propositions:

(137) Let us consider a natural number q. Suppose $2 \leq q$. Then

- (i) there exists a strict graph G such that G is a thin spider of order q, and
- (ii) there exists a strict graph G such that G is a thick spider of order q.

(138) Let us consider objects a, z, and a set X. Then $z \in \{a\} \times X$ if and only if there exists an object j such that $j \in X$ and $z = \langle a, j \rangle$.

(139) Let us consider natural numbers q, i. Suppose G is a thin spider of order q and $i \in \operatorname{Seg} q$. Then
  - (i) for every element x of G such that $x = \langle 0, i \rangle$ holds $N(x) = \{0\} \times (\operatorname{Seg} q \setminus \{i\}) \cup \{1\} \times \{i\}$, and
  - (ii) for every element x of G such that $x = \langle 1, i \rangle$ holds $N(x) = \{0\} \times \{i\}$.

(140) Let us consider natural numbers q, i. Suppose G is a thick spider of order q and $i \in \operatorname{Seg} q$. Then
  - (i) for every element x of G such that $x = \langle 0, i \rangle$ holds $N(x) = \{0\} \times (\operatorname{Seg} q \setminus \{i\}) \cup \{1\} \times (\operatorname{Seg} q \setminus \{i\})$, and
  - (ii) for every element x of G such that $x = \langle 1, i \rangle$ holds $N(x) = \{0\} \times (\operatorname{Seg} q \setminus \{i\})$.

(141) Let us consider natural numbers q, i, a function g from $\mathbb{N}$ into $\mathbb{N}$, and finite subsets S, S_0 of $\mathbb{N}$. Suppose $i \in \operatorname{Seg} q$ and $S = \operatorname{Seg} q$ and $S_0 = \operatorname{Seg} q \setminus \{i\}$. Then $\sum (g \restriction S) = \sum (g \restriction S_0) + g(i)$. The theorem is a consequence of (4).

(142) Let us consider a natural number q, a subset A of $\operatorname{Seg} q$, and an object a. Suppose $\Omega_G = \{0, 1\} \times \operatorname{Seg} q$ and ($a = 0$ or $a = 1$). Then $\{a\} \times A \subseteq \Omega_G$.

(143) Let us consider natural numbers q, i, functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$, and a finite subset S of $\mathbb{N}$. Suppose G is a thin spider of order q and $i \in \operatorname{Seg} q$ and (for every natural number j such that $j \in \operatorname{Seg} q$ holds $f(\langle 0, j \rangle) = f_u(j)$ and $f(\langle 1, j \rangle) = f_v(j)$) and $S = \operatorname{Seg} q$. Then
  - (i) for every element x of G such that $x = \langle 0, i \rangle$ holds $f(N(x)) + f_u(i) = \sum (f_u \restriction S) + f_v(i)$, and
  - (ii) for every element y of G such that $y = \langle 1, i \rangle$ holds $f(N(y)) = f_u(i)$.

(144) Let us consider natural numbers q, i, functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$, and a finite subset S of $\mathbb{N}$. Suppose G is a thick spider of order q and $i \in \operatorname{Seg} q$ and (for every natural number j such that $j \in \operatorname{Seg} q$ holds $f(\langle 0, j \rangle) = f_u(j)$ and $f(\langle 1, j \rangle) = f_v(j)$) and $S = \operatorname{Seg} q$. Then
  - (i) for every element x of G such that $x = \langle 0, i \rangle$ holds $f(N(x)) + f_u(i) + f_v(i) = \sum (f_u \restriction S) + \sum (f_v \restriction S)$, and
  - (ii) for every element y of G such that $y = \langle 1, i \rangle$ holds $f(N(y)) + f_u(i) = \sum (f_u \restriction S)$.

(145) Let us consider a natural number q. Then $q+1 \leq 2 \cdot \lceil \frac{q+1}{2} \rceil$ and $2 \cdot \lceil \frac{q+1}{2} \rceil \leq q+2$.

The scheme *NatCases* deals with a unary predicate $\mathcal{P}$ and two unary functors $\mathcal{F}$, $\mathcal{H}$ from the natural numbers into the natural numbers, and states that there exists a function g from $\mathbb{N}$ into $\mathbb{N}$ such that for every natural number i, if $\mathcal{P}[i]$, then $g(i) = \mathcal{F}(i)$, and if not $\mathcal{P}[i]$, then $g(i) = \mathcal{H}(i)$. Now we state the propositions:

(146) Let us consider a natural number q. Suppose $\Omega_G = \{0, 1\} \times \operatorname{Seg} q$. Then there exists a natural number i such that $i \in \operatorname{Seg} q$ and ($v = \langle 0, i \rangle$ or $v = \langle 1, i \rangle$).

(147) Let us consider an object a, and a finite set X. Then $\overline{\overline{\{a\} \times X}} = \overline{\overline{X}}$.

(148) Let us consider a natural number q. Suppose (G is a thin spider of order q or G is a thick spider of order q) and (for every elements x, y of G and

every natural numbers i, j such that $i \in \operatorname{Seg} q$ and $j \in \operatorname{Seg} q$ and $i \neq j$ and $x = \langle 0, i\rangle$ and $y = \langle 0, j\rangle$ holds $f(N(x)) \neq f(N(y))$) and (for every elements x, y of G and every natural numbers i, j such that $i \in \operatorname{Seg} q$ and $j \in \operatorname{Seg} q$ and $\langle x, y\rangle \in E_G$ and $x = \langle 0, i\rangle$ and $y = \langle 1, j\rangle$ holds $f(N(x)) \neq f(N(y))$). Then f is additive.

(149) Let us consider a natural number q, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose $\Omega_G = \{0, 1\} \times \operatorname{Seg} q$. Then there exists f such that for every natural number j such that $j \in \operatorname{Seg} q$ holds $f(\langle 0, j\rangle) = f_u(j)$ and $f(\langle 1, j\rangle) = f_v(j)$.

(150) Let us consider natural numbers q, k, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose $\Omega_G = \{0, 1\} \times \operatorname{Seg} q$ and (for every natural number j such that $j \in \operatorname{Seg} q$ holds $f(\langle 0, j\rangle) = f_u(j)$ and $f(\langle 1, j\rangle) = f_v(j)$) and (for every natural number j such that $j \in \operatorname{Seg} q$ holds $1 \leq f_u(j)$ and $f_u(j) \leq k$ and $1 \leq f_v(j)$ and $f_v(j) \leq k$). Then f is k-labeling.

(151) Let us consider a natural number q. Suppose $2 \leq q$ and G is a thin spider of order q. Then $\lceil \frac{q+1}{2} \rceil \leq \eta(G)$.

(152) Let us consider a natural number q. Suppose $2 \leq q$ and G is a thin spider of order q. Then $\eta(G) \leq \lceil \frac{q+1}{2} \rceil$.

(153) Let us consider a natural number q. Suppose $2 \leq q$ and G is a thin spider of order q. Then $\eta(G) = \lceil \frac{q+1}{2} \rceil$. The theorem is a consequence of (152) and (151).

(154) Let us consider a natural number q. Suppose $2 \leq q$ and G is a thick spider of order q. Then $\lceil \frac{q+1}{2} \rceil \leq \eta(G)$.

(155) Let us consider a natural number q. Suppose $2 \leq q$ and G is a thick spider of order q. Then $\eta(G) \leq \lceil \frac{q+1}{2} \rceil$.

(156) Let us consider a natural number q. Suppose $2 \leq q$ and G is a thick spider of order q. Then $\eta(G) = \lceil \frac{q+1}{2} \rceil$. The theorem is a consequence of (155) and (154).

10. Suns

Let G be a graph, m be a natural number, B be a set, and R be a relation. We say that G is the sun of order m on B with base R if and only if

(Def. 31) $\Omega_G = \{0\} \times B \cup \{1\} \times \operatorname{Seg} m$ and for every objects x, y holds $\langle x, y\rangle \in E_G$ iff (there exist natural numbers i, j such that $x = \langle 0, i\rangle$ and $y = \langle 0, j\rangle$ and $\langle i, j\rangle \in R$) or (there exist natural numbers i, j such that $i \in \operatorname{Seg} m$ and ($j = i$ or $j = \operatorname{pred}_m(i)$) and (($x = \langle 0, i\rangle$ and $y = \langle 1, j\rangle$) or ($x = \langle 1, j\rangle$ and $y = \langle 0, i\rangle$))).

Now we state the propositions:

(157) Let us consider a natural number m, a finite subset B of $\mathbb{N}$, and a relation R. Suppose $3 \leq m$ and $\operatorname{Seg} m \subseteq B$ and $R \subseteq B \times B$ and (for every objects i, j such that $\langle i, j\rangle \in R$ holds $\langle j, i\rangle \in R$) and (for every object i holds $\langle i, i\rangle \notin R$). Then there exists a strict graph G such that G is the sun of order m on B with base R.

(158) Let us consider natural numbers m, j. Suppose $2 \leq m$ and $j \in \operatorname{Seg} m$. Then $\mathrm{succ}_m(j) \neq j$ and $\mathrm{pred}_m(j) \neq j$.

From now on B denotes a subset of $\mathbb{N}$, R denotes a relation. Now we state the propositions:

(159) Let us consider natural numbers m, j, and an element x of G. Suppose G is the sun of order m on B with base R and $j \in \operatorname{Seg} m$ and $x = \langle 1, j \rangle$. Then $N(x) = \{\langle 0, j \rangle, \langle 0, \mathrm{succ}_m(j) \rangle\}$.

(160) Let us consider natural numbers m, i, and an element x of G. Suppose G is the sun of order m on B with base R and $R \subseteq B \times B$ and $x = \langle 0, i \rangle$. Then $N(x) = \{0\} \times R^{\circ}\{i\} \cup \{1\} \times (\{j \in \mathbb{N} : i \in \operatorname{Seg} m \text{ and } (j = i \text{ or } j = \mathrm{pred}_m(i))\})$.

(161) Let us consider natural numbers m, i. Then

- (i) if $i \in \operatorname{Seg} m$, then $\{j \in \mathbb{N} : i \in \operatorname{Seg} m \text{ and } (j = i \text{ or } j = \mathrm{pred}_m(i))\} = \{i, \mathrm{pred}_m(i)\}$, and
- (ii) if $i \notin \operatorname{Seg} m$, then $\{j \in \mathbb{N} : i \in \operatorname{Seg} m \text{ and } (j = i \text{ or } j = \mathrm{pred}_m(i))\} = \emptyset$.

(162) Let us consider a natural number m, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose G is the sun of order m on B with base R and $R \subseteq B \times B$ and $3 \leq m$ and $\operatorname{Seg} m \subseteq B$ and (for every natural number j such that $j \in B$ holds $f(\langle 0, j \rangle) = f_u(j)$) and (for every natural number j such that $j \in \operatorname{Seg} m$ holds $f(\langle 1, j \rangle) = f_v(j)$). Then

- (i) for every natural number j and every element y of G such that $j \in \operatorname{Seg} m$ and $y = \langle 1, j \rangle$ holds $f(N(y)) = f_u(j) + f_u(\mathrm{succ}_m(j))$, and
- (ii) for every natural number i and every element x of G and every finite subset R_i of $\mathbb{N}$ such that $i \in \operatorname{Seg} m$ and $x = \langle 0, i \rangle$ and $R_i = R^{\circ}\{i\}$ holds $f(N(x)) = \sum (f_u \restriction R_i) + f_v(i) + f_v(\mathrm{pred}_m(i))$, and
- (iii) for every natural number i and every element x of G and every finite subset R_i of $\mathbb{N}$ such that $i \notin \operatorname{Seg} m$ and $x = \langle 0, i \rangle$ and $R_i = R^{\circ}\{i\}$ holds $f(N(x)) = \sum (f_u \restriction R_i)$.

(163) Let us consider a natural number m, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose G is the sun of order m on B with base R. Then there exists f such that (for every natural number j such that $j \in B$ holds $f(\langle 0, j \rangle) = f_u(j)$) and (for every natural number j such that $j \in \operatorname{Seg} m$ holds $f(\langle 1, j \rangle) = f_v(j)$).

(164) Let us consider a natural number m. Suppose G is the sun of order m on B with base R. Then there exists a natural number i such that ($i \in B$ and $v = \langle 0, i \rangle$) or ($i \in \operatorname{Seg} m$ and $v = \langle 1, i \rangle$).

(165) Let us consider natural numbers m, k, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose G is the sun of order m on B with base R and (for every natural number j such that $j \in B$ holds $f(\langle 0, j \rangle) = f_u(j)$ and $1 \leq f_u(j)$ and $f_u(j) \leq k$) and (for every natural number j such that $j \in \operatorname{Seg} m$ holds $f(\langle 1, j \rangle) = f_v(j)$ and $1 \leq f_v(j)$ and $f_v(j) \leq k$). Then f is k-labeling.

(166) Let us consider a natural number m. Suppose G is the sun of order m on B with base R and (for every elements x, y of G and every natural numbers i, j such that $\langle i, j \rangle \in R$ and $x = \langle 0, i \rangle$ and $y = \langle 0, j \rangle$ holds $f(N(x)) \neq f(N(y))$) and (for every elements x, y of G and every natural numbers i, j such that

$i \in \operatorname{Seg} m$ and ($j = i$ or $j = \operatorname{pred}_m(i)$) and $x = \langle 0, i\rangle$ and $y = \langle 1, j\rangle$ holds $f(N(x)) \neq f(N(y))$). Then f is additive.

Let m be a natural number. The functor K_m yielding a relation on $\operatorname{Seg} m$ is defined by

(Def. 32) for every natural numbers i, j holds $\langle i, j\rangle \in it$ iff $i \in \operatorname{Seg} m$ and $j \in \operatorname{Seg} m$ and $i \neq j$.

Let m be a natural number. The functor W_m yielding a relation on $\{0\} \cup \operatorname{Seg} m$ is defined by

(Def. 33) for every natural numbers i, j holds $\langle i, j\rangle \in it$ iff ($i \in \operatorname{Seg} m$ and $j \in \operatorname{Seg} m$ and ($j = \operatorname{succ}_m(i)$ or $i = \operatorname{succ}_m(j)$)) or ($i = 0$ and $j \in \operatorname{Seg} m$) or ($j = 0$ and $i \in \operatorname{Seg} m$).

Now we state the propositions:

(167) Let us consider a relation R, and objects i, j. Then $j \in R^\circ\{i\}$ if and only if $\langle i, j\rangle \in R$.

(168) Let us consider natural numbers m, i. Suppose $i \in \operatorname{Seg} m$. Then $(K_m)^\circ\{i\} = \operatorname{Seg} m \setminus \{i\}$.

(169) Let us consider natural numbers m, i. Suppose $i \in \operatorname{Seg} m$. Then $(C_m)^\circ\{i\} = \{\operatorname{succ}_m(i), \operatorname{pred}_m(i)\}$.

(170) Let us consider natural numbers m, i. Suppose $i \in \operatorname{Seg} m$. Then $(W_m)^\circ\{i\} = \{\operatorname{succ}_m(i), \operatorname{pred}_m(i)\} \cup \{0\}$ and $(W_m)^\circ\{0\} = \operatorname{Seg} m$.

11. Cycle Suns and Wheel Suns

Now we state the propositions:

(171) Let us consider a natural number i. Then
  (i) if i is odd, then $i + 1$ is even, and
  (ii) if i is even, then $i + 1$ is odd, and
  (iii) 1 is odd, and
  (iv) 2 is even.

(172) Let us consider natural numbers m, i. Suppose $i \in \operatorname{Seg} m$. Then
  (i) if $i < m$, then $\operatorname{succ}_m(i) = i + 1$, and
  (ii) if $i = m$, then $\operatorname{succ}_m(i) = 1$, and
  (iii) if $1 < i$, then $\operatorname{pred}_m(i) = i - 1$, and
  (iv) if $i = 1$, then $\operatorname{pred}_m(i) = m$.

(173) Let us consider natural numbers m, i, a finite set R_i, and an element x of G. Suppose G is the sun of order m on B with base R and $R \subseteq B \times B$ and $3 \leq m$ and $i \in \operatorname{Seg} m$ and $R_i = R^\circ\{i\}$ and $x = \langle 0, i\rangle$. Then $\deg x = \overline{\overline{R_i}} + 2$.

(174) Let us consider a natural number m, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose $4 \leq m$ and (for every natural number i holds (if i is odd, then $f_u(i) = 2$) and (if i is not odd, then $f_u(i) = 1$)) and (for every natural number i holds (if $i = 1$ and m is odd, then $f_v(i) = 2$) and (if not ($i = 1$ and m is odd), then $f_v(i) = 1$)). Then

(i) for every natural number i such that $i \in \operatorname{Seg} m$ holds $f_u(\mathrm{succ}_m(i)) + f_u(\mathrm{pred}_m(i)) + f_v(i) + f_v(\mathrm{pred}_m(i)) \neq f_u(\mathrm{succ}_m(\mathrm{succ}_m(i))) + f_u(\mathrm{pred}_m(\mathrm{succ}_m(i))) + f_v(\mathrm{succ}_m(i)) + f_v(\mathrm{pred}_m(\mathrm{succ}_m(i)))$, and

(ii) for every natural number i such that $i \in \operatorname{Seg} m$ holds $4 \leq f_u(\mathrm{succ}_m(i)) + f_u(\mathrm{pred}_m(i)) + f_v(i) + f_v(\mathrm{pred}_m(i))$ and $f_u(\mathrm{succ}_m(i)) + f_u(\mathrm{pred}_m(i)) + f_v(i) + f_v(\mathrm{pred}_m(i)) \leq 7$, and

(iii) for every natural number i such that $i \in \operatorname{Seg} m$ and m is even holds $f_u(\mathrm{succ}_m(i)) + f_u(\mathrm{pred}_m(i)) + f_v(i) + f_v(\mathrm{pred}_m(i)) = 4$ or $f_u(\mathrm{succ}_m(i)) + f_u(\mathrm{pred}_m(i)) + f_v(i) + f_v(\mathrm{pred}_m(i)) = 6$, and

(iv) for every natural number j such that $j \in \operatorname{Seg} m$ holds $f_u(j) + f_u(\mathrm{succ}_m(j)) < f_u(\mathrm{succ}_m(j)) + f_u(\mathrm{pred}_m(j)) + f_v(j) + f_v(\mathrm{pred}_m(j))$ and $f_u(j) + f_u(\mathrm{succ}_m(j)) < f_u(\mathrm{succ}_m(\mathrm{succ}_m(j))) + f_u(\mathrm{pred}_m(\mathrm{succ}_m(j))) + f_v(\mathrm{succ}_m(j)) + f_v(\mathrm{pred}_m(\mathrm{succ}_m(j)))$.

(175) Let us consider a natural number m. Suppose $4 \leq m$ and G is the sun of order m on $\operatorname{Seg} m$ with base C_m. Then $\eta(G) = 2$.

(176) Let us consider a natural number m, and functions f_u, f_v from $\mathbb{N}$ into $\mathbb{N}$. Suppose $4 \leq m$ and G is the sun of order m on $\{0\} \cup \operatorname{Seg} m$ with base W_m and (for every natural number j such that $j \in \{0\} \cup \operatorname{Seg} m$ holds $f(\langle 0, j \rangle) = f_u(j)$) and (for every natural number j such that $j \in \operatorname{Seg} m$ holds $f(\langle 1, j \rangle) = f_v(j)$). Then

(i) for every natural number i and every element x of G such that $i \in \operatorname{Seg} m$ and $x = \langle 0, i \rangle$ holds $f(N(x)) = f_u(\mathrm{succ}_m(i)) + f_u(\mathrm{pred}_m(i)) + f_u(0) + f_v(i) + f_v(\mathrm{pred}_m(i))$ and $\deg x = 5$, and

(ii) for every element x of G such that $x = \langle 0, 0 \rangle$ holds for every finite subset S of $\mathbb{N}$ such that $S = \operatorname{Seg} m$ holds $f(N(x)) = \sum (f_u \restriction S)$, and

(iii) for every natural number j and every element y of G such that $j \in \operatorname{Seg} m$ and $y = \langle 1, j \rangle$ holds $f(N(y)) = f_u(j) + f_u(\mathrm{succ}_m(j))$.

(177) Let us consider a function f_u from $\mathbb{N}$ into $\mathbb{N}$, a natural number n, and a finite subset S of $\mathbb{N}$. Suppose (for every natural number i holds (if i is odd, then $f_u(i) = 2$) and (if i is not odd, then $f_u(i) = 1$)) and $S = \operatorname{Seg} n$. Then

(i) if n is even, then $2 \cdot \sum (f_u \restriction S) = 3 \cdot n$, and

(ii) if n is odd, then $2 \cdot \sum (f_u \restriction S) = 3 \cdot n + 1$.

(178) Let us consider a natural number i. Suppose $i \in \operatorname{Seg} 5$. Then $i = 1$ or $i = 2$ or $i = 3$ or $i = 4$ or $i = 5$.

(179) Let us consider a natural number m. Suppose $4 \leq m$ and G is the sun of order m on $\{0\} \cup \operatorname{Seg} m$ with base W_m. Then there exist elements u_1, u_2 of G such that $u_1 = \langle 0, 1 \rangle$ and $u_2 = \langle 0, 2 \rangle$ and u_1 and u_2 are adjacent and $\deg u_1 = 5$ and $\deg u_2 = 5$.

(180) Let us consider a function g from $\mathbb{N}$ into $\mathbb{N}$, a natural number n, and finite subsets S, S_1 of $\mathbb{N}$. Suppose $S = \operatorname{Seg} n$ and $S_1 = \operatorname{Seg}(n + 1)$. Then $\sum (g \restriction S_1) = \sum (g \restriction S) + g(n + 1)$. The theorem is a consequence of (4).

(181) Let us consider a natural number m. Suppose $4 \leq m$ and G is the sun of order m on $\{0\} \cup \operatorname{Seg} m$ with base W_m. Then $\eta(G) = 2$.

12. Complete Suns

Now we state the propositions:

(182) Let us consider natural numbers x, t, c. Suppose $x = 3 \cdot t + c$ and $c < 3$. Then $x \operatorname{div} 3 = t$.

(183) Let us consider a natural number c. Suppose $c < 3$. Then $c = 0$ or $c = 1$ or $c = 2$.

(184) Let us consider a natural number c. Suppose $c < 6$. Then $c = 0$ or $c = 1$ or $c = 2$ or $c = 3$ or $c = 4$ or $c = 5$.

(185) Let us consider a natural number x. Then $(x \operatorname{div} 3) + ((x + 2) \operatorname{div} 3) + ((x + 4) \operatorname{div} 3) = x + 1$.

Let m, i be natural numbers. The functor $\mathfrak{q}_m(i)$ yielding a natural number is defined by the term

$$\text{(Def. 34)} \quad \begin{cases} 1, & \text{if } i = 1, \\ 2 \cdot (i \mathbin{\dot{-}} 1), & \text{if } i \neq 1 \text{ and } i \leq (m \operatorname{div} 2) + 1, \\ 3 + 2 \cdot (m \mathbin{\dot{-}} i), & \text{otherwise.} \end{cases}$$

Let m, i be natural numbers. The functor $\mathrm{ku}_m(i)$ yielding a natural number is defined by the term

$$\text{(Def. 35)} \quad \begin{cases} \left\lceil \frac{m+2}{3} \right\rceil, & \text{if } (m \operatorname{mod} 3) = 2 \text{ and } i = ((m + 1) \operatorname{div} 2) + 1, \\ (\mathfrak{q}_m(i) \operatorname{div} 3) + 1, & \text{otherwise.} \end{cases}$$

Let m, i be natural numbers. The functor $\mathrm{ka}_m(i)$ yielding a natural number is defined by the term

$$\text{(Def. 36)} \quad \begin{cases} (\mathfrak{q}_m(i) + 2) \operatorname{div} 3, & \text{if } i = 1 \text{ or } (m \operatorname{div} 2) + 2 \leq i, \\ \left\lceil \frac{m+2}{3} \right\rceil \mathbin{\dot{-}} 1, & \text{if not } (i = 1 \text{ or } (m \operatorname{div} 2) + 2 \leq i) \text{ and } (m \operatorname{mod} 6) = 2 \text{ and } i = ((m + 1) \operatorname{div} 2) + 1, \\ (\mathfrak{q}_m(i) + 4) \operatorname{div} 3, & \text{otherwise.} \end{cases}$$

Let m, i be natural numbers. The functor $\mathrm{kv}_m(i)$ yielding a natural number is defined by the term

$$\text{(Def. 37)} \quad (\left\lceil \frac{m+2}{3} \right\rceil + 1) \mathbin{\dot{-}} \mathrm{ka}_m(i).$$

Now we state the propositions:

(186) Let us consider natural numbers x, t, c. Suppose $x = 3 \cdot t + c$ and $c < 3$. Then $x \operatorname{mod} 3 = c$.

(187) Let us consider natural numbers x, t, c. Suppose $x = 2 \cdot t + c$ and $c < 2$. Then $x \operatorname{div} 2 = t$ and $x \operatorname{mod} 2 = c$.

(188) Let us consider a natural number m. Then there exists a natural number t such that $((m \operatorname{mod} 6) = 0$ and $m = 6 \cdot t$ and $(m \operatorname{mod} 3) = 0$ and m is even) or $((m \operatorname{mod} 6) = 1$ and $m = 6 \cdot t + 1$ and $(m \operatorname{mod} 3) = 1$ and m is odd) or $((m \operatorname{mod} 6) = 2$ and $m = 6 \cdot t + 2$ and $(m \operatorname{mod} 3) = 2$ and m is even) or $((m \operatorname{mod} 6) = 3$ and $m = 6 \cdot t + 3$ and $(m \operatorname{mod} 3) = 0$ and m is odd) or $((m \operatorname{mod} 6) = 4$ and $m = 6 \cdot t + 4$ and $(m \operatorname{mod} 3) = 1$ and m is even) or $((m \operatorname{mod} 6) = 5$ and $m = 6 \cdot t + 5$ and $(m \operatorname{mod} 3) = 2$ and m is odd).

(189) Let us consider a natural number m. Suppose $3 \leq m$. Then

(i) $\left\lceil \frac{m+2}{3} \right\rceil = (m + 4) \operatorname{div} 3$, and

(ii) $3 \cdot \lceil \frac{m+2}{3} \rceil \leq m+4$, and
(iii) $m+4 < 3 \cdot \lceil \frac{m+2}{3} \rceil + 3$, and
(iv) $2 \leq \lceil \frac{m+2}{3} \rceil$, and
(v) $\lceil \frac{m+2}{3} \rceil < m$, and
(vi) $2 \cdot (m \operatorname{div} 2) \leq m$, and
(vii) $m \leq 2 \cdot (m \operatorname{div} 2) + 1$, and
(viii) $1 \leq m \operatorname{div} 2$, and
(ix) if m is even, then $m = 2 \cdot (m \operatorname{div} 2)$ and $((m+1) \operatorname{div} 2)+1 = (m \operatorname{div} 2)+1$, and
(x) if m is odd, then $m = 2 \cdot (m \operatorname{div} 2) + 1$ and $((m+1) \operatorname{div} 2) + 1 = (m \operatorname{div} 2) + 2$, and
(xi) $3 \leq ((m+1) \operatorname{div} 2) + 1$, and
(xii) if $(m \bmod 6) = 2$, then m is even and $(m \bmod 3) = 2$, and
(xiii) if $(m \bmod 3) = 2$ and m is even, then $(m \bmod 6) = 2$, and
(xiv) if $(m \bmod 3) = 2$ and m is odd, then $(m \bmod 6) = 5$.

(190) Let us consider natural numbers m, i. Suppose $3 \leq m$. Then
(i) if $i = 1$, then $\mathfrak{q}_m(i) = 1$, and
(ii) if $2 \leq i$ and $i \leq (m \operatorname{div} 2) + 1$, then $\mathfrak{q}_m(i) + 2 = 2 \cdot i$, and
(iii) if $(m \operatorname{div} 2) + 2 \leq i$ and $i \leq m$, then $\mathfrak{q}_m(i) + 2 \cdot i = 2 \cdot m + 3$, and
(iv) if not $((m \bmod 3) = 2$ and $i = ((m+1) \operatorname{div} 2) + 1)$, then $\mathrm{ku}_m(i) = (\mathfrak{q}_m(i) \operatorname{div} 3) + 1$, and
(v) if $(m \bmod 3) = 2$ and $i = ((m+1) \operatorname{div} 2) + 1$, then $\mathrm{ku}_m(i) = \lceil \frac{m+2}{3} \rceil$, and
(vi) if $i = 1$ or $(m \operatorname{div} 2) + 2 \leq i$, then $\mathrm{ka}_m(i) = (\mathfrak{q}_m(i) + 2) \operatorname{div} 3$, and
(vii) if $2 \leq i$ and $i \leq (m \operatorname{div} 2) + 1$ and $(m \bmod 6) = 2$ and $i = ((m+1) \operatorname{div} 2) + 1$, then $\mathrm{ka}_m(i) = \lceil \frac{m+2}{3} \rceil \mathbin{\dot{-}} 1$, and
(viii) if $2 \leq i$ and $i \leq (m \operatorname{div} 2) + 1$ and not $((m \bmod 6) = 2$ and $i = ((m+1) \operatorname{div} 2) + 1)$, then $\mathrm{ka}_m(i) = (\mathfrak{q}_m(i) + 4) \operatorname{div} 3$.

(191) Let us consider a natural number m. Suppose $3 \leq m$. Then
(i) if $(m \bmod 6) = 2$, then $m+3 = 2 \cdot \lceil \frac{m+2}{3} \rceil + ((m+2) \operatorname{div} 3)$ and $m+1 = \lceil \frac{m+2}{3} \rceil + ((m+1) \operatorname{div} 3) + ((m \mathbin{\dot{-}} 1) \operatorname{div} 3)$, and
(ii) if $(m \bmod 6) = 5$, then $m+2 = \lceil \frac{m+2}{3} \rceil + ((m+2) \operatorname{div} 3) + ((m+3) \operatorname{div} 3)$, and
(iii) if $(m \bmod 6) = 0$ or $(m \bmod 6) = 4$, then $m = ((m+1) \operatorname{div} 3) + ((m+4) \operatorname{div} 3) + ((m \mathbin{\dot{-}} 1) \operatorname{div} 3)$, and
(iv) if $(m \bmod 6) = 1$ or $(m \bmod 6) = 3$, then $m+1 = ((m+2) \operatorname{div} 3) + ((m+3) \operatorname{div} 3) + (m \operatorname{div} 3)$.

(192) Let us consider natural numbers m, i. Suppose $3 \leq m$ and $i \in \operatorname{Seg} m$. Then $\mathfrak{q}_m(i) + 2 = \mathrm{ka}_m(i) + \mathrm{ka}_m(\mathrm{pred}_m(i)) + \mathrm{ku}_m(i)$.

(193) Let us consider a natural number x. Then $(x+3) \operatorname{div} 3 = (x \operatorname{div} 3) + 1$. The theorem is a consequence of (182).

(194) Let us consider natural numbers m, i. Suppose $3 \leq m$ and $i \in \operatorname{Seg} m$. Then
(i) $1 \leq \mathfrak{q}_m(i)$, and
(ii) $\mathfrak{q}_m(i) \leq m$, and
(iii) $1 \leq \mathrm{ku}_m(i)$, and

    (iv) $\mathrm{ku}_m(i) \leq \lceil \frac{m+2}{3} \rceil$, and
    (v) $1 \leq \mathrm{ka}_m(i)$, and
    (vi) $\mathrm{ka}_m(i) \leq \lceil \frac{m+2}{3} \rceil$, and
    (vii) $\mathrm{kv}_m(i) + \mathrm{ka}_m(i) = \lceil \frac{m+2}{3} \rceil + 1$, and
    (viii) $1 \leq \mathrm{kv}_m(i)$, and
    (ix) $\mathrm{kv}_m(i) \leq \lceil \frac{m+2}{3} \rceil$.

(195) Let us consider natural numbers a, b. Then $2 \cdot a \neq 2 \cdot b + 1$. The theorem is a consequence of (171).

(196) Let us consider natural numbers m, i, j. Suppose $3 \leq m$ and $i \in \mathrm{Seg}\, m$ and $j \in \mathrm{Seg}\, m$ and $\mathfrak{q}_m(i) = \mathfrak{q}_m(j)$. Then $i = j$.

(197) Let us consider a natural number m. Suppose $3 \leq m$. Then $\mathfrak{q}_m(m) = 3$ and $\mathrm{ku}_m(m) = 2$.

(198) Let us consider a natural number m. Suppose $3 \leq m$ and G is the sun of order m on $\mathrm{Seg}\, m$ with base K_m. Then $\eta(G) = \lceil \frac{m+2}{3} \rceil$.

(199) Let us consider natural numbers a, b. Suppose $0 < b$. Then $\lceil \frac{a}{b} \rceil = \lceil a/b \rceil$.

(200) Let us consider a natural number m. Suppose $3 \leq m$. Then
    (i) there exists a strict graph G such that G is the sun of order m on $\mathrm{Seg}\, m$ with base K_m, and
    (ii) there exists a strict graph G such that G is the sun of order m on $\mathrm{Seg}\, m$ with base C_m, and
    (iii) there exists a strict graph G such that G is the sun of order m on $\{0\} \cup \mathrm{Seg}\, m$ with base W_m.

ACKNOWLEDGEMENT: This document has the layout of a paper in *Formalized Mathematics*. It is generated by `miz2tex.py` from the Mizar article `acn.miz` and has not been published.

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